

PROJECT REPORT

Multilingual Conversational AI Assistant

Internship Project – Project 6

1. Abstract

The Multilingual Conversational AI Assistant is a retrieval-based conversational AI application designed to support customer-service interactions across multiple languages. It supports English, Hindi, Marathi and Gujarati while preserving conversational continuity during language switches. The system combines language detection, mixed-language detection, intent detection, conversation memory, multilingual embeddings, FAISS retrieval, intent-aware re-ranking and context-aware response generation.

2. Problem Statement

Single-language chatbots may fail when users switch languages, mix languages within one message, or ask follow-up questions without repeating the topic. This project addresses these requirements through an integrated multilingual conversational pipeline.

3. Objectives

The objectives are automatic language identification, support for at least three additional languages, mixed-language handling, conversation context retention, cross-lingual intent understanding, semantic retrieval and consistent responses.

4. Architecture

User Message → Language Detection → Mixed-Language Detection → Conversation Memory → Intent Detection → Multilingual Embedding → FAISS Retrieval → Intent-Aware Re-ranking → Context-Aware Response.

5. Technology Stack

Python 3.11, Streamlit, Sentence Transformers, paraphrase-multilingual-MiniLM-L12-v2, FAISS, LangDetect, Unicode script detection, NumPy, Pandas and PyTorch.

6. Multilingual Processing

The multilingual embedding model places semantically equivalent text from different languages into a shared vector space. Language detection combines LangDetect with Unicode script signals, which is useful for Gujarati and code-mixed Devanagari/Latin inputs.

7. Conversation Memory

Previous user and assistant turns are retained in conversation state. Switching from English to Hindi, Marathi or Gujarati therefore does not reset the conversation.

8. Intent Detection

The system handles customer-service intents including order status, delivery time, cancellation, refund, return, address change, payment issue and product information.

9. Cross-Lingual Retrieval

FAISS performs similarity search over multilingual embeddings. Multiple candidates are retrieved and then re-ranked using the detected intent to reduce incorrect matches among closely related categories.

10. Mixed-Language Handling

The system processes examples such as 'Mera order ■■■ ■■ ■■■■ ■■■', which combines Latin-script words with Hindi written in Devanagari.

11. User Interface

The Streamlit interface provides chat history, multilingual responses, a clear-conversation control and an AI Analysis panel containing language, intent, confidence, mixed-language status and retrieval information.

12. Functional Testing

Validation covered English, Hindi, Marathi, Gujarati, language switching, mixed-language input, multiple customer-service intents, FAISS index creation, cross-lingual retrieval and the Streamlit application. The implemented functional tests passed during final validation.

13. Results

The final prototype demonstrates multilingual conversations, language switching, response-language matching, intent changes, cross-lingual semantic retrieval, mixed-language handling and conversational continuity. These are functional validation results rather than a statistically rigorous benchmark.

14. Limitations

Very short messages may be difficult to classify. Hindi and Marathi share Devanagari script, making short inputs potentially ambiguous. Similarity scores vary between languages and the knowledge base is intentionally controlled and domain-specific.

15. Future Scope

Future enhancements include more languages, stronger language identification, multilingual generative LLM responses, persistent conversation storage, larger evaluation datasets, automated metrics and cloud deployment.

16. Conclusion

The project demonstrates a practical open-source architecture for multilingual customer-service conversations. The combination of language detection, conversation memory, intent detection, multilingual embeddings, FAISS retrieval and intent-aware re-ranking provides a strong foundation for cross-lingual conversational AI.

17. Validation Screenshot 1



मेरा ऑर्डर कब आएगा?



डिलीवरी में सामान्यतः 3-5 कार्यदिवस लगते हैं।

>

AI Analysis



आणि address बदलायचा असेल तर?



ऑर्डर डिस्पॅच होण्यापूर्वी तुम्ही डिलिव्हरीचा पत्ता बदलू शकता.

>

AI Analysis



Can I cancel it?



You can cancel the order if it has not yet been dispatched.

>

AI Analysis



મને રિફંડ જોઈએ છે.



પાન ઓર્ડર ૨૬ અથવા પરત કર્યા પછી રિફંડની વિનંતી કરી શકાય છે.

>

AI Analysis

Type your message in English, Hindi, Marathi, or Gujarati...

>

17. Validation Screenshot 2

Assistant Information

Supported Languages

GB English

IN Hindi

IN Marathi

IN Gujarati

AI Components

Sentence Transformers

FAISS

LangDetect

Structured Conversation Memory

Clear Conversation

Deploy

 **Multilingual Conversational AI Assistant**

This chatbot demonstrates multilingual conversational AI with:

- Automatic language detection
- Language switching
- Conversation memory
- Intent detection
- Cross-lingual semantic retrieval
- Context-aware responses
- GB English
- IN Hindi
- IN Marathi
- IN Gujarati

 Mera order अभी तक नहीं आया

 डिलीवरी में सामान्यतः 3-5 कार्यदिवस लगते हैं।

>  AI Analysis

Type your message in English, Hindi, Marathi, or Gujarati...